Mahindra University
Global Thinkers. Engaged Leaders.
ÉCOLE CENTRALE SCHOOL OF ENGINEERING

Mathematics-1 (MA 1101)
Tutorial Sheet-1
Topics: Continuity, Differentiability, Indeterminate Forms and Taylor Series

***Problems to be worked out in tutorial**

†Assignment Problems

Pb 1) * Assume that $f : [0, 2] \to \mathbb{R}$ is a continuous function and $f(0) = f(2)$. Prove that there exist $x_1,\ x_2 \in [0, 2]$ such that $x_2 - x_1 = 1$ and $f(x_2) = f(x_1)$.

Pb 2) Suppose f is a continuous real valued function on the interval $[0, 1]$. Show that $\int_0^1 x^2 f(x)\mathrm{d}x = \frac{1}{3}f(c)$ for some $c \in [0, 1]$.

Pb 3) * Let $f : [a, b] \to \mathbb{R} \setminus \mathbb{Q}$ be continuous. Prove that f must be constant.

Pb 4) $f : \mathbb{R} \to \mathbb{R}$ is a continuous function with period 2. Prove that the graphs of $y = f(x)$ and $y = f(x + 1)$ intersect somewhere.

Pb 5) * $f : \mathbb{R} \to \mathbb{R}$ is continuous and even, while $g : \mathbb{R} \to \mathbb{R}$ is continuous and odd. If $f(3) = 0$, show that for some $c \in [-3, 3]$, $f(c) = g(c)$.

Pb 6) * Assume that f is integrable and non-negative on $[a, b]$. If $\int_a^b f(x)dx = 0$, prove that $f(x) = 0$ at each point of continuity of f.

Pb 7) † Assume that f is continuous on $[a, b]$ and $\int_a^b f(x)g(x)dx = 0$ for every function g that is continuous on $[a, b]$. Prove that $f(x) = 0$ for all x in $[a, b]$.

Pb 8) * Prove that if f is continuous on $[a, b]$, differentiable on (a, b), and if $f(a) = f(b) = 0$, then for real α there is an $x \in (a, b)$ such that $\alpha f(x) + f'(x) = 0$.

Pb 9) † Assume that f is continuous on $[a, b]$, $a > 0$ and differentiable on (a, b). Show that if $\frac{f(a)}{a} = \frac{f(b)}{b}$, then there is $x_0 \in (a, b)$ such that $x_0 f'(x_0) = f(x_0)$.

Pb 10) * Assume that $a_0, a_1, ..., a_n$ are real numbers such that $\frac{a_0}{n+1} + \frac{a_1}{n} + ... + a_n = 0$. Prove that the polynomial $p(x) = a_0x^n + a_1x^{n-1} + ... + a_n$ has at least one root in $(0, 1)$.

Pb 11) Using Mean value theorem, prove that if $x > 0$

i) * $\ln(1 + x) < x$ ii) $\sin x < x$ iii) $\mathrm{e}^x > 1 + x + \frac{x^2}{2}$

Pb 12) Prove that $x^3 - 3x + c$ has at most one root in [0,1] no matter what c may be.

Pb 13) Suppose f is continuous on [-2,2], differentiable on (-2,2). If $f(-2) = -1$ and $f'(x) \geq 3$ for all x. What is the least possible value for $f(2)$.

Pb 14) Suppose $f'(x) = 0$ for all $x \in [a, b]$, then $f(x)$ is a constant function on (a, b).

Pb 15) † Suppose that a function f is twice differentiable on $[0, 4]$ and that $f(1) = f(2) = 0$, and $f(3) = 1$. Prove that

i) $f'(a) = \frac{1}{2}$ for some $a \in (0,4)$.

ii) $f'(b) \geq \frac{1}{2}$ for some $b \in (0,4)$.

Pb 16) * Let $f : [0,1] \mapsto \mathbb{R}$ be a function twice differentiable, such that $f^{(2)}$ is continuous. Assume that $f(0) = f'(0) = f'(1) = 0$ and $f(1) = 1$. Prove that there exists $c \in [0,1]$ such that $|f''(c)| \geq 4$.

Pb 17) * Find $(a,b) \in \mathbb{R}^2$ such that $\lim_{x \to 0} \left(\frac{e^x}{\ln(1+x)} + \frac{a}{x} + b \right) = 0$.

Pb 18) Compute $\lim_{n \to \infty} n \times \left(\left(1 + \frac{2}{n}\right)^n - \left(1 + \frac{1}{n}\right)^{2n} \right)$.

Pb 19) † Let $f : \mathbb{R} \mapsto \mathbb{R}$ be defined as $f(x) = e^{-\frac{1}{x}}$ if $x > 0$, $f(x) = 0$ if $x \leq 0$.

i) Show that f is infinitely differentiable on $(-\infty, 0)$ and on $(0, +\infty)$.

ii) Show that, $\forall n \in \mathbb{N}$, there exists a polynomial function p_n such that $f^{(n)}(x) = \frac{p_n(x)}{x^{2n}} e^{-\frac{1}{x}}$ if $x > 0$.

iii) Prove that $\forall n \in \mathbb{Z}$, $\lim_{x \to 0} \frac{e^{-\frac{1}{x}}}{x^n} = 0$ using the Taylor's approximation of e^x.

iv) Prove that $\lim_{x \to 0} f^{(n)}(x) = 0$.

v) Prove that if g is a continuous function, differentiable on $\mathbb{R} \setminus \{0\}$ and such that $\lim_{x \to 0\,,\, x \neq 0} g'(x) = l$, then g is differentiable at 0 and $g'(0) = l$.

vi) Use *i)*, *iv)* and *v)* to conclude that f is infinitely differentiable on $\mathbb{R}$ and that $\forall n \in \mathbb{N}$, $f^{(n)}(0) = 0$.

vii) What is the n-th order Taylor polynomial at point 0 of f ?

Pb 20) * Let $f : \mathbb{R} \to \mathbb{R}$ be $n+1$ times differentiable on $\mathbb{R}$. Prove that for every $x \in \mathbb{R}$ there is $\theta \in (0,1)$ such that $f(x) = f(0) + xf'(x) - \frac{x^2}{2} f''(x) + \ldots\ldots + (-1)^{n+1} \frac{x^n}{n!} f^{(n)}(x) + (-1)^{n+2} \frac{x^{n+1}}{(n+1)!} f^{(n+1)}(\theta x)$.